

Regression

What is supervised learning

Regression Math

Solar Wind Prediction (Jupyter Notebook)

Meet Your Instructors

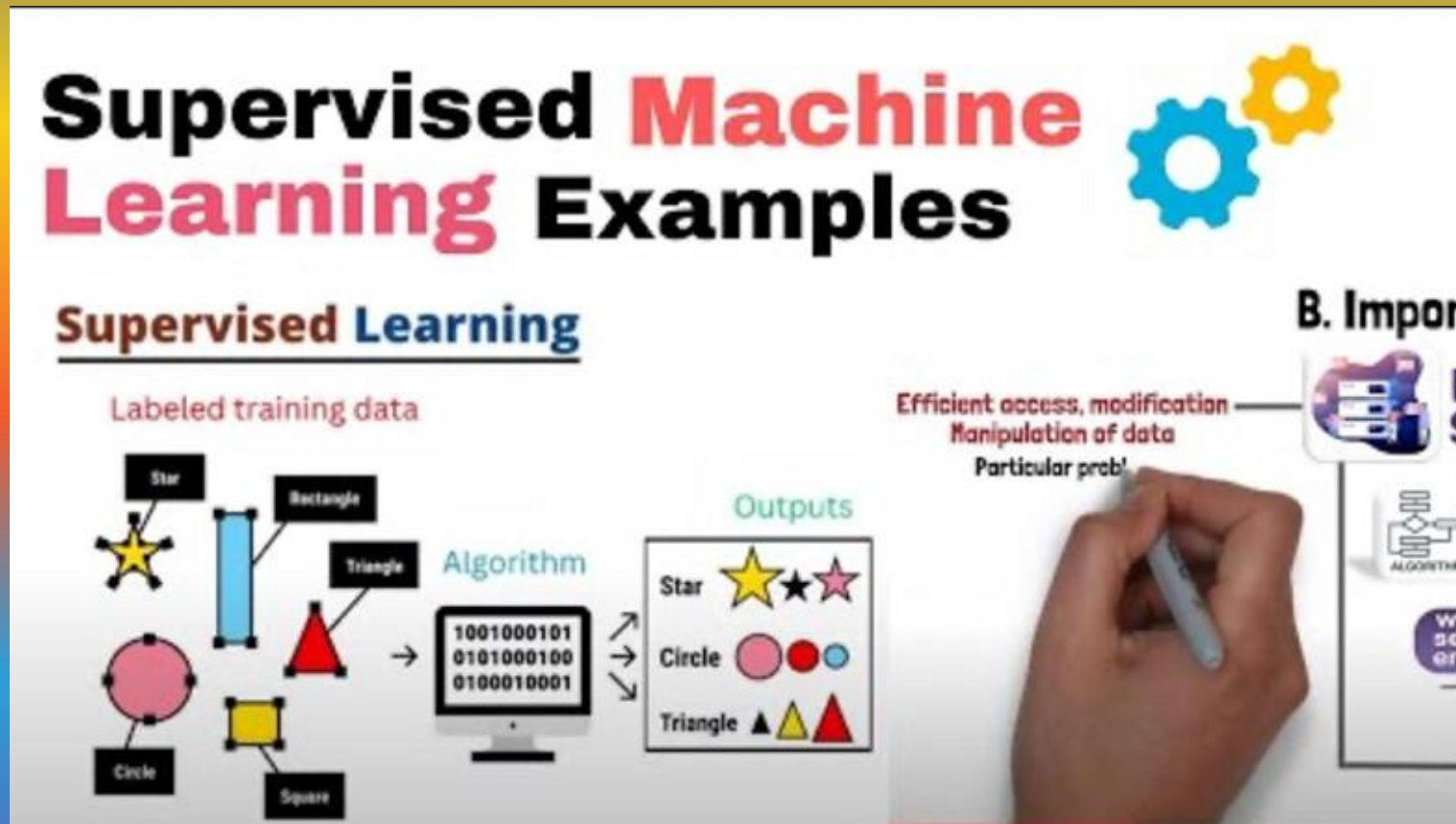
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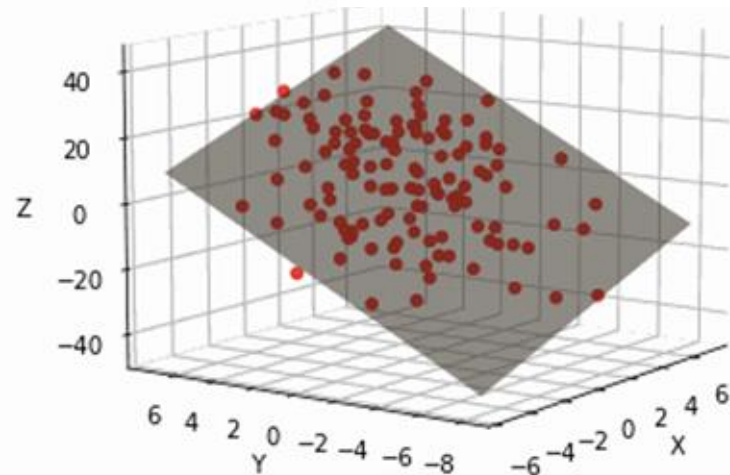
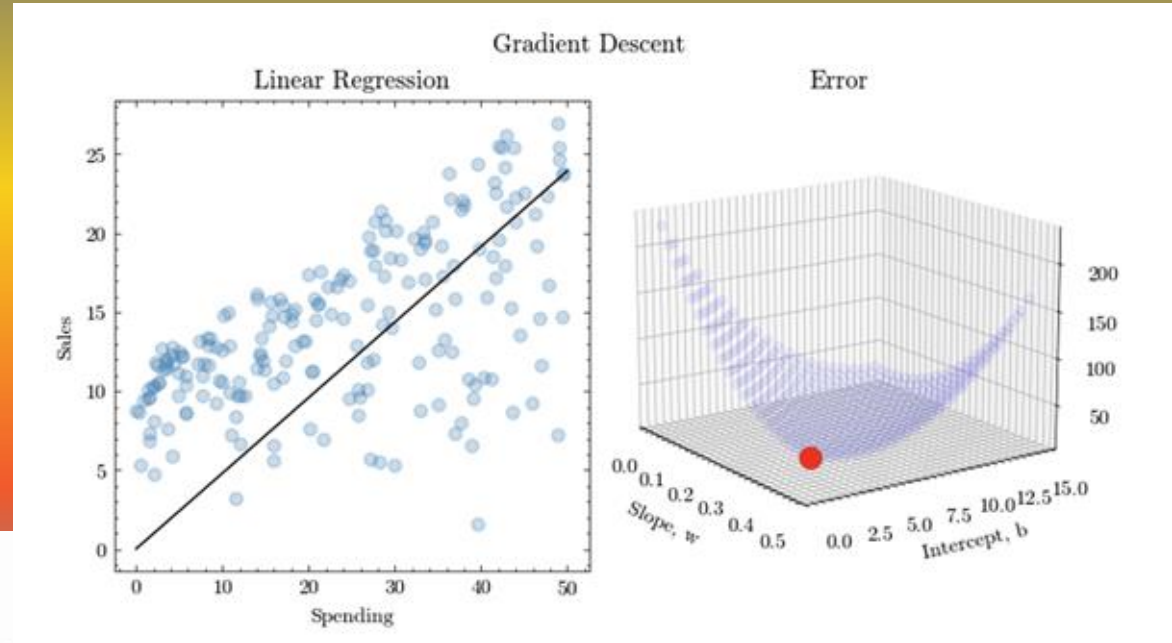
What is Supervised Learning?



https://youtu.be/Ps0y6w4cD_U?si=rwjY1DIky6otGe7X

Regression

- Predicts numeric value
- Types:
 - Polynomial
 - Logistic
 - Linear
- Applications:
 - Finance
 - Economics
 - Healthcare



Statistics

Statistical Models rely on mathematical foundations for finding and explaining the relationships between variables, typically to find correlations and predictions from data. SMs are also used for descriptive summaries of statistical information.

Diagram illustrating the linear regression equation $y = mx + b$ with labels:

- Dependent Variable**: y
- Indendent Variable**: x
- Where line crosses the y-axis**: b
- Y- Intercept**: b
- Coefficient, Rate and Slope of line**: m

Machine Learning

Machine Learning is a computer science perspective on modeling data with a focus on algorithmic methods and model skill. The machine learning approach offers the possibility of automatically learning rules from the data that map inputs to outputs. The approach of learning directly from the data can lead to discovering the hidden knowledge of relationships within data.

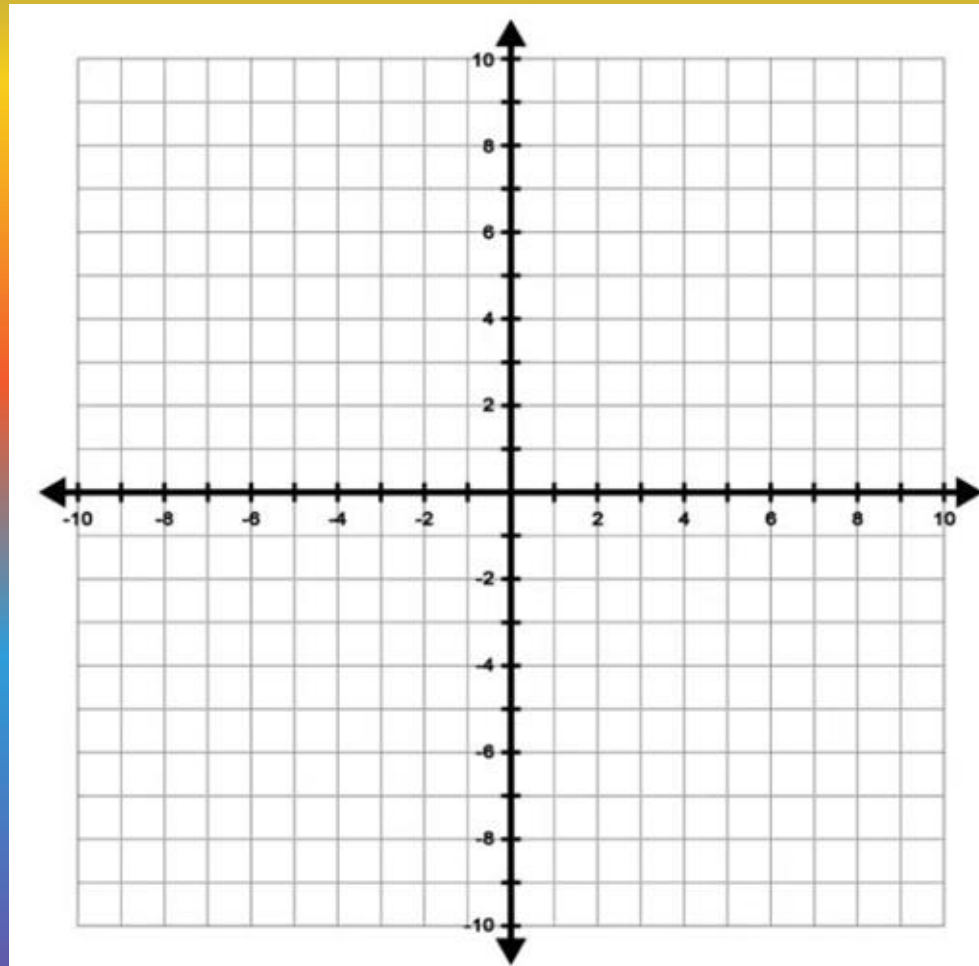
Diagram illustrating the machine learning equation $y' = b + w_1x_1$ with labels:

- Prediction**: y'
- Bias**: b
- Weight**: w_1
- Feature value**: x_1

Calculated from training

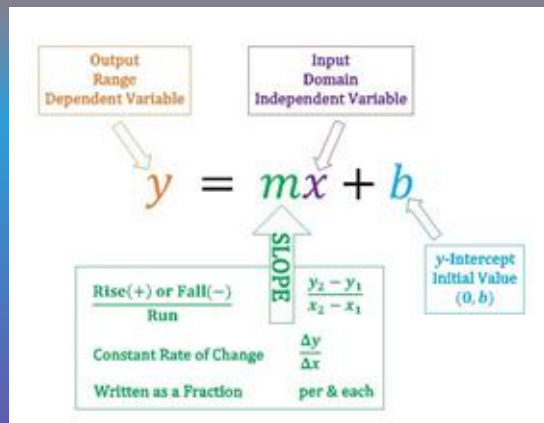
The Line (Two Points)

- Slope
- Point-Slope
- Slope-Intercept Form
- Line Predictions



Given the following two points, find and plot the line. $(-4, -2)$, $(2, -8)$

Use your new equation to predict the y value when $x = -8$.



The Line (> Two Points)

- **Mean:** The average value of a dataset, found by summing all values and dividing by the number of observations.
- **Deviations:** The differences between each data point and the mean of the dataset.
- **Sum of Squares:** The total of the squared deviations from the mean, measuring overall variability in the data.
- **Sum of Cross Deviations:** The sum of the products of deviations of two variables from their respective means, used to measure how they vary together.
- **Least Squares Fit Line:** The line that minimizes the total squared differences between observed values and predicted values in a linear regression.
- **Residuals:** The differences between the actual observed values and the values predicted by the model.
- **Mean Square Error:** The average of the squared residuals, representing how far predictions are from actual values on average.

$$\text{Least Squares Fit: } \hat{y} = m\hat{x} + b$$

Given the following three points, find the line that fits them: $(-4, -2)$, $(2, -8)$, $(3, -7)$.

Mean:

$$\bar{x} = \frac{1}{n} \sum x_i$$

$$\bar{y} = \frac{1}{n} \sum y_i$$

Deviations:

$x_i - \bar{x}$	$y_i - \bar{y}$

Sum of Squares:

$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
$\sum (x_i - \bar{x})^2 =$	$\sum (y_i - \bar{y})^2 =$

Sum of Cross Deviations: $(x_i - \bar{x})(y_i - \bar{y})$

$\sum (x_i - \bar{x})(y_i - \bar{y}) =$

Slope:

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} =$$

Intercept:

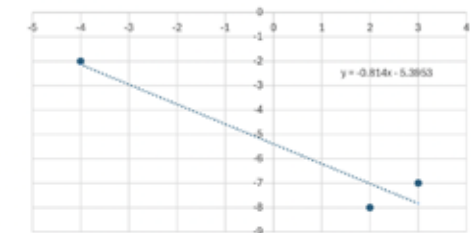
$$b = \bar{y} - m\bar{x} =$$

Final Line:

$$\hat{y} = m\hat{x} + b =$$

Predictions:

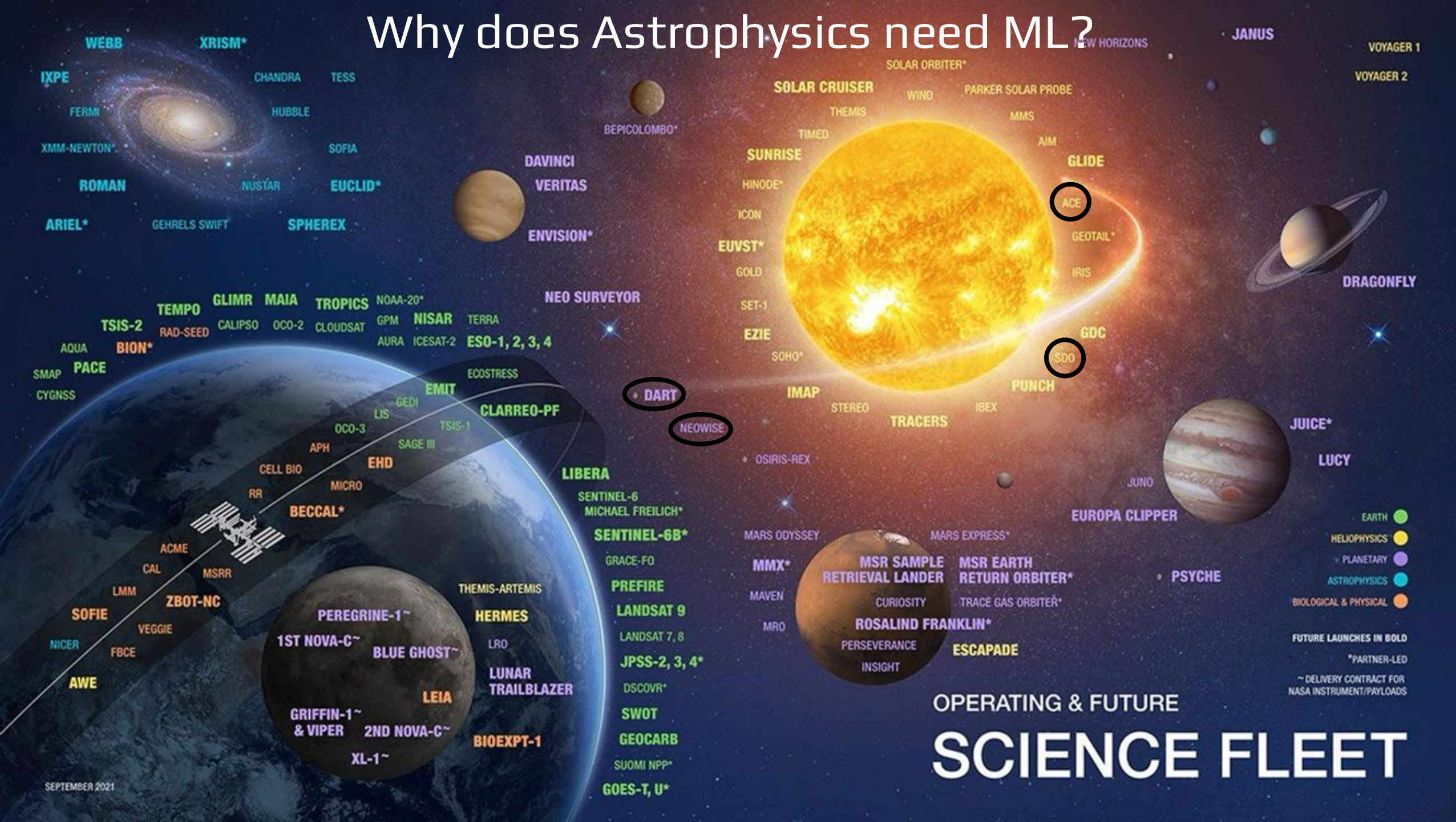
x_i	y_i	$\hat{y}_i = m\hat{x}_i + b$	$(\hat{x}_i, \hat{y}_i)$	Residuals: $e_i = y_i - \hat{y}_i$
-4	-2			
2	-8			
3	-7			



Mean Square Error:

$$\frac{1}{n} \sum (y_i - \hat{y}_i)^2 =$$

Why does Astrophysics need ML?

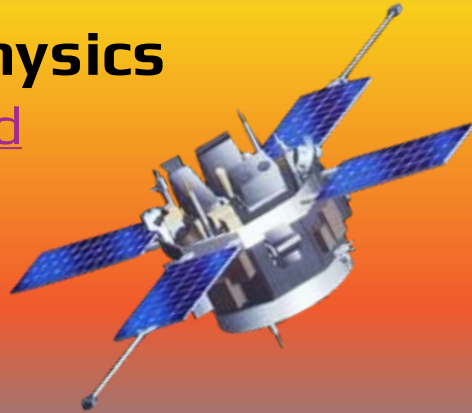


The Data

Heliophysics

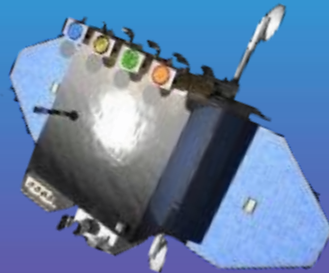
- ACE EPAM Solar Wind

- February 2013
- 5-minute cadence
- 8065 indices



- SDO AIA Images (131 A)

- June 2011
- 12-second cadence
- 1520 events



Astrophysics

- NASA Infrared Telescope Facility (IRFT)

- SMASSII and Tholen
 - Small Body Database
 - NASA Infra
 - 800,000 asteroids



ML Transition: ACE Example

- **Inputs (x) / Outputs (y):** Inputs (x) are the variables used to make predictions, and outputs (y) are the values the model is trying to predict.
- **Training vs Testing:** Training data is used to fit the model, while testing data is used to evaluate how well the model performs on unseen data.
- **Loss Function = MSE:** The loss function MSE measures the average squared difference between predicted and actual values, guiding the model to minimize error.
- **Generalization:** Generalization refers to how well a model performs on new, unseen data rather than just the data it was trained on.
- **Overfitting (conceptual):** Overfitting occurs when a model learns patterns specific to the training data (including noise) and performs poorly on new data.

Guided Example: Linear Regression for Solar Wind Prediction (ACE EPAM)

This notebook walks through linear regression in two ways:

1. **Statistics view:** fit a line and evaluate fit (error).
2. **Machine Learning view:** treat linear regression as a model that is trained, tested, and used to predict on unseen data.

Dataset (CSV):
ace_epam_2023-02.csv, hosted on S3

Machine Learning Concepts Covered (in markdown as you requested)

- Inputs (x) / Outputs (y)
- Training vs testing
- Loss function: Mean Squared Error (MSE)
- Generalization (performance on unseen data)
- Overfitting (conceptual)

Note: This example uses time as the input feature and one EPAM channel as the target.
For real forecasting, you'd often use multiple features and/or time-lagged inputs.

```
# If needed, install dependencies (uncomment)
# !pip install pandas numpy matplotlib scikit-learn

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

plt.rcParams["figure.figsize"] = (8, 4)
```

1) Load the dataset from S3

We will load the CSV directly from the URL. Then we will:

- inspect columns
- identify a time column (if present)
- select one numeric EPAM channel as the target y

```
1) DATA_URL = "https://indiajacksonphd.s3-us-east-1.amazonaws.com/NRAD/ace_epam_2023-02.csv"
df = pd.read_csv(DATA_URL)
df.head()

2) df.shape, df.columns.tolist()[1:30] # show first ~30 column names
```

2) Inputs (x) and Outputs (y)

Inputs (x)

For this guided example, we use time as the single input feature.

- If the dataset already has a time column, we parse it.
- If it does not, we create an index-based time variable.

Output (y)

We choose one numeric EPAM channel to predict.

Below, you can set:

- TIME_COL: (or let the notebook auto-detect)
- TARGET_COL: (or let the notebook pick a numeric column automatically)

```
# --- Configure these if you want to force specific columns ---
TIME_COL = None # e.g., "time", "datetime", "timestamp"
TARGET_COL = None # e.g., "P2", "P2", "Flux", etc.

# --- Auto-detect a time column if not provided ---
if TIME_COL is None:
    for candidate in ["time", "datetime", "date", "timestamp", "Time", "DATE", "Datetime"]:
        if candidate in df.columns:
            TIME_COL = candidate
            break

TIME_COL
```

Click Here: <http://149.165.147.51/hub/login>